José **Ribeiro**

Cybersecurity Operator · Virtualization & Cyber Defense

Portugal

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"If my name is on it, I need to be proud of it."

Summary

Cyber Defense Operator with hands-on experience in cybersecurity operations, virtualization, and cyber range environments within a military context. Skilled in Windows Server and Linux system administration, system hardening, and participation in cybersecurity training exercises. Experienced in phishing simulation campaigns, security awareness initiatives, and threat scenario development. Strong technical focus on operational security and infrastructure resilience.

Education

Instituto Politécnico de Beja — *Portugal*

Master's Degree in Cybersecurity Engineering and Cybercrime — *2026 – 2028*

ISTEC Lisboa — *Portugal*

CTeSP - Cybersecurity — *2024 – 2026*

IUM – Instituto Universitário Militar, Escola de Ciberdefesa — *Portugal*

Digital Forensics Analyst Course — *Feb. 2022 – Feb. 2023*

- Digital forensics analysis, incident response and cybersecurity training

Work Experience

Cyber Defense Operations Command – Portuguese Armed Forces General Staff — *Portugal*

Systems Administration & Cybersecurity Operations — *Mar. 2026 – Present*

- Administration and secure operation of critical infrastructure services, including Active Directory, DNS, DHCP, AD CS, Root CA, and Exchange
- Management and hardening of Windows Server and Linux systems, ensuring secure configuration, patching, access control, and service resilience
- Administration of backup and recovery operations using Veeam Backup & Replication, supporting system availability, data protection, and disaster recovery
- Administration of virtualization and converged infrastructure platforms, including VMware ESXi and Cisco HyperFlex
- Implementation and maintenance of certificate services, trust chains, and authentication-related controls to support secure internal communications
- Monitoring, troubleshooting, and securing services to ensure availability, integrity, confidentiality, and compliance with cyberdefence requirements
- Support for infrastructure protection through vulnerability mitigation, baseline security enforcement, log analysis, and operational security best practices
- Participation in SOC activities, including monitoring, log analysis, and identification of potential security incidents

Cyber Defense Operations Command – Portuguese Armed Forces General Staff — *Portugal*

Virtualization Administrator & Phishing Simulation — *Mar. 2023 – Present*

- Management of virtualization platforms and cyber range environments for advanced cybersecurity training
- Design and deployment of virtual infrastructures and custom network topologies
- Administration of Windows Server and Linux systems with a focus on system hardening and security best practices
- Development and execution of phishing simulation and security awareness campaigns
- Creation of threat scenarios for cyber defense training exercises (blue/red team)

Portuguese Air Force — *Portugal*

Military Service — *Nov. 2021 – Present*

- Operational military service with assignment to cyber defense-related activities

Certifications

RUMUS

Check Point Security Administrator R81.20 (CCSA) *2025*

GALILEO

Administration and Configuration Windows Server 2022 *2025*

SCRUMstudy

Scrum Fundamentals Certified (SFC) *2025*

Technical Skills

Operating Systems	Windows Server, Linux
System Administration	Active Directory, DNS, DHCP, User and Permission Management, System Hardening
Virtualization	VMware ESXi, OS Virtualization, Cyber Range Environments, Virtual Infrastructure Design
Backup & Recovery	Veeam Backup & Replication, Backup Management, Restore Operations, Disaster Recovery
Networking	TCP/IP, DNS, DHCP, VLANs, Network Segmentation
Cybersecurity & SOC	Log Analysis, Security Monitoring, Incident Detection, SIEM (QRadar), Threat Analysis
Digital Forensics	FTK Imager, Autopsy, File Recovery, Memory Analysis (Volatility), Evidence Handling
Security & Analysis Tools	Wireshark, Volatility
Professional Skills	Technical Documentation, Incident Analysis, Security Awareness Training, Operational Discipline

Cybersecurity Projects

Internal Cyber Defense Exercise Environment

VMware, Windows Server, Linux, pfSense, Trellix

Cyber Range Infrastructure for Cyber Defense Exercises (CyberDex 2024/2026)

- Designed and deployed a full-scale cyber range infrastructure to support internal blue/red team exercises.
- Implemented enterprise-like environments including Active Directory, DNS, DHCP, Exchange, Windows and Linux workstations.
- Configured pfSense firewall and endpoint security agents (McAfee/Trellix) to simulate realistic defensive conditions.
- Prepared vulnerable systems and attack surfaces to enable controlled red team intrusion scenarios.

Detection and Data Exfiltration Prevention

Wazuh, pfSense, Suricata

Security Monitoring and Log Correlation Architecture

- Designed a security architecture integrating SIEM and network-based detection mechanisms.
- Implemented log collection and correlation using Wazuh for centralized visibility.
- Configured Suricata IDS/IPS on pfSense to detect anomalous traffic and potential exfiltration attempts.
- Developed monitoring strategies to improve detection capabilities and strengthen security posture.

Security Awareness and Testing Framework

GoPhish, Moodle, SMTP

Phishing Simulation Platform (Open-Source)

- Developed a phishing simulation infrastructure using open-source tools for user awareness testing.
- Integrated GoPhish with Moodle and an internal SMTP server for campaign management.
- Demonstrated a cost-effective approach for organizations to assess user susceptibility to phishing.

Risk-Based User Selection Model

Python, Data Analysis

Adaptive Phishing Campaign System with AI-Based Targeting

- Designed a model to dynamically select phishing targets based on user risk profiles.
- Incorporated factors such as role criticality, data sensitivity, and prior behavior in previous campaigns.
- Improved the effectiveness of awareness campaigns through intelligent targeting mechanisms.

Access Control and Deception Mechanisms

Honeypots, Behavioral Analysis

Secure Authentication Architecture with Behavioral Detection and Honeypots

- Designed a conceptual architecture combining physical login systems with behavioral monitoring.
- Integrated deception mechanisms to detect unauthorized access attempts.
- Explored methods to correlate authentication anomalies with potential threat activity.

System Protection and Access Control

SELinux, AppArmor

Linux System Hardening using SELinux and AppArmor

- Implemented mandatory access control policies to restrict system-level operations.
- Configured SELinux and AppArmor profiles to protect critical files and services.
- Improved system resilience against privilege escalation and unauthorized access.

Wi-Fi Security Assessment

Kali Linux, Aircrack-ng

WPA2-PSK Wireless Attack and Defense Analysis

- Executed capture and cracking of WPA2 handshakes.
- Analyzed weaknesses in wireless configurations and password policies.
- Proposed mitigation strategies to strengthen wireless network security.

Incident Investigation and Evidence Analysis

Volatility, FTK Imager, Autopsy, PhotoRec

Digital Forensics Analysis of Compromised Systems

- Participated in forensic investigations of compromised systems.
- Performed memory analysis, file recovery, and metadata examination.
- Supported evidence handling and reporting for incident analysis.

AD, PKI, Web Services, SSO

Windows Server 2022, IIS, LDAP, FTP

Enterprise Infrastructure with Secure Services and Web Integration

- Deployed enterprise infrastructure including Active Directory, DNS, DHCP, and Exchange services.
- Implemented SSL certificates and secure file sharing mechanisms.
- Developed an intranet with LDAP authentication and Single Sign-On integration.

Resilience, Replication and Backup

Hyper-V, VMware

High Availability Virtualized Infrastructure

- Designed infrastructure with domain services, GPOs, file servers, and virtualization platforms.
- Implemented replication, live migration, and backup/recovery strategies.
- Ensured service availability and disaster recovery readiness.

Offensive Security and Intelligence Gathering

Tor, ProxyChains, Nmap

Anonymous Reconnaissance Platform with MITRE ATT&CK Mapping

- Developed a platform for anonymous reconnaissance using Tor and proxy chaining.
- Integrated scanning tools and vulnerability discovery techniques.
- Mapped results to the MITRE ATT&CK framework and generated structured reports.

Detection, Deception and Zero Trust

pfSense, Zeek, Darktrace, QRadar, OpenVPN, Cloudflare

Advanced Cyber Defense Architecture (Red October Simulation)

- Designed a layered defense architecture against advanced persistent threats.
- Implemented SIEM, network monitoring, and anomaly detection capabilities.
- Integrated Zero Trust access through OpenVPN and Cloudflare.
- Developed deception mechanisms and network segmentation strategies.

NATO CCDCOE Cyber Exercise

Windows Server, ADCS

Locked Shields Participation (Blue Team)

- Participated in the annual Locked Shields exercise as part of a blue team.
- Responsible for defending Windows infrastructure, including critical services such as ADCS.
- Handled incident response, service hardening, and system recovery under attack conditions.